Flipkart Logistics Optimization Project Report

1. Introduction

This project focuses on analyzing and optimizing delivery logistics for Flipkart using SQL and Power BI. The goal is to improve delivery efficiency, reduce delays, and enhance overall operational performance.

2. Objectives

- Analyze delivery delays
- Optimize delivery routes
- Evaluate warehouse performance
- Measure delivery agent efficiency
- Build interactive dashboards for insights

3. Tools Used

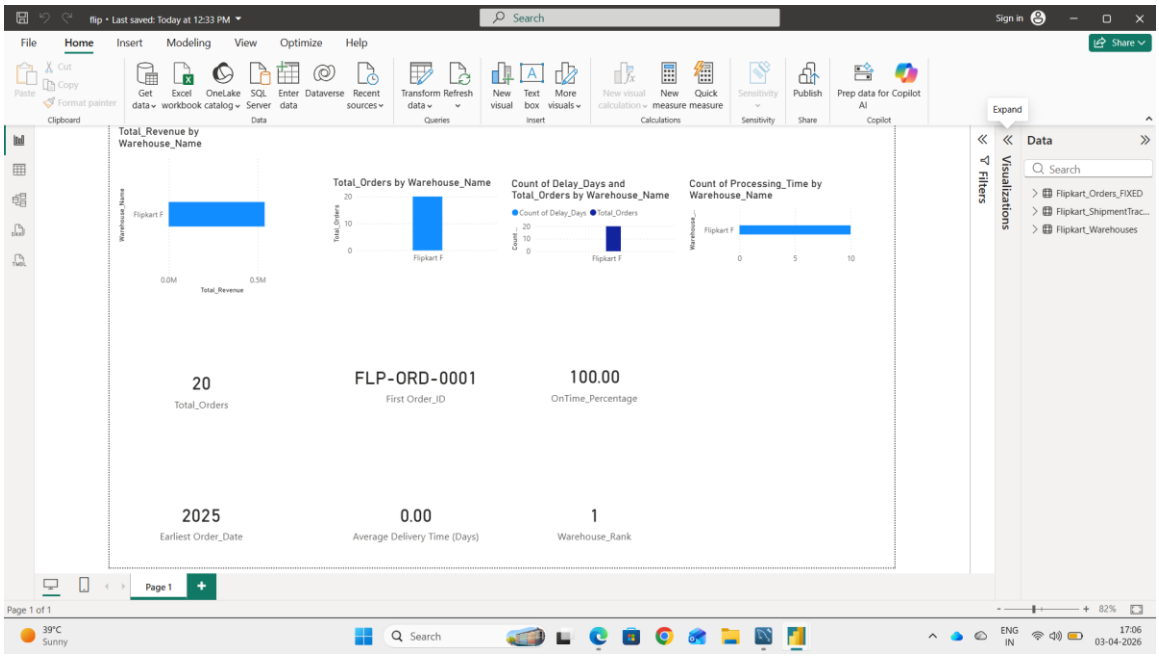
- SQL (Data Analysis)
- Power BI (Visualization)
- Excel/CSV (Dataset)

4. Data Description

The dataset includes:
- Orders data (Order_ID, dates, status)
- Routes data (distance, travel time, traffic delay)
- Delivery agents data (efficiency, speed)
- Shipment tracking (delay reasons, checkpoints)
- Warehouses data

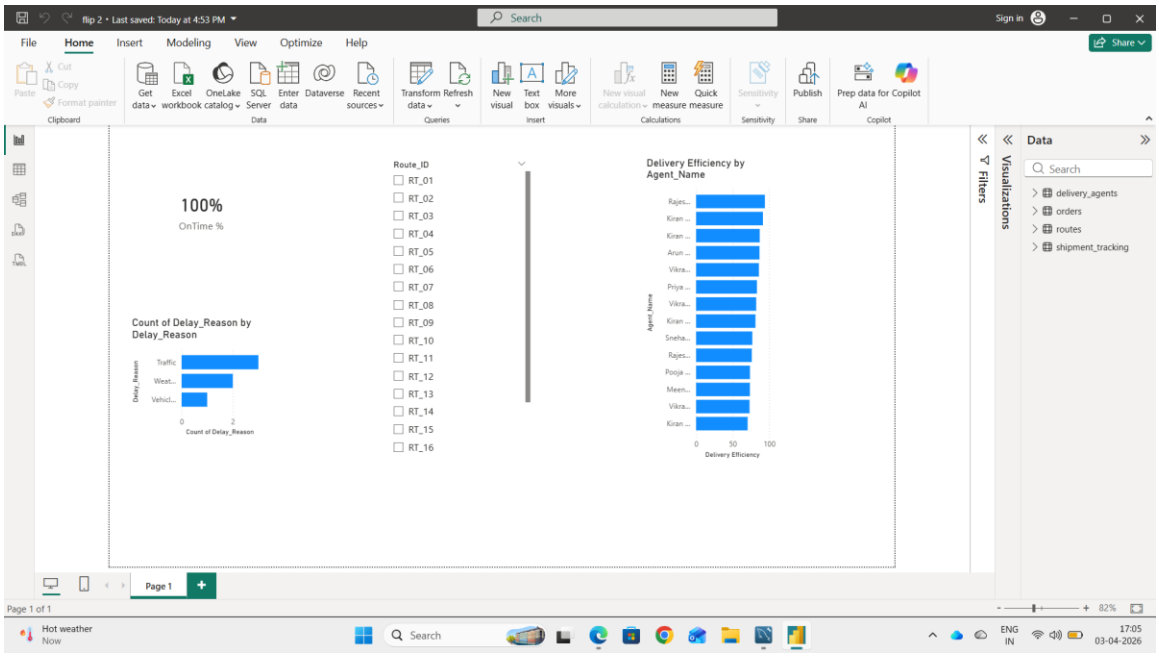
5. Key Analysis Performed

- Data cleaning and handling missing values
- Delay calculation using DATEDIFF
- Route efficiency analysis
- Warehouse performance comparison
- Agent efficiency evaluation
- Shipment delay tracking



6. Key KPIs

- On-Time Delivery Percentage
- Average Delivery Delay
- Traffic Delay Analysis
- Route Efficiency
- Delivery Agent Efficiency



7. Dashboard Insights

- Most delays are caused by traffic conditions
- Certain routes have higher delivery delays
- Top-performing agents maintain high efficiency
- Warehouse performance varies based on processing time

8. Conclusion

The project successfully identifies key factors affecting delivery performance. By optimizing routes and improving agent efficiency, delivery operations can be enhanced significantly. Power BI dashboards provide clear and actionable insights for decision-making.